

Safety Formation Control for Multirobot Systems

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ABSTRACT

This project investigates safe formation control for multi-robot systems using a distributed formation controller combined with Control Barrier Functions. A distributed consensus-based controller is first employed to generate nominal control inputs that drive the agents toward a desired formation using only local neighbor information. To guarantee collision avoidance, a CBF-based safety filter is introduced to minimally modify the nominal control input while enforcing safety constraints between agents and surrounding obstacles. To improve scalability and reduce conservativeness, a Relaxed Distributed Control Barrier Function formulation is adopted. The proposed framework is validated through collision-prone simulation scenarios, including a position-swapping task and formation control in environments containing static obstacles and a region of disturbance. Simulation results demonstrate that the robots successfully achieve the desired formation while maintaining the minimum safety distance throughout the motion.

In addition, ongoing work focuses on developing an event-triggered communication strategy to reduce communication overhead and computational burden in large-scale multi-robot systems. The objective is to guarantee lower-bounded inter-event times while preserving safety even when agents do not communicate continuously.

Key words: control barrier function, distributed formation control, collision avoidance, event-triggered control, multi-robot systems, safety-critical control.

1 INTRODUCTION

Recently, multi-robot systems have attracted increasing attention due to their applications in drone shows, autonomous transportation, surveillance, and cooperative robotics [1–4]. In these applications, multiple robots operate in a shared environment and coordinate with one another to achieve a common objective, the most common one is forming and maintaining a desired formation [5].

However, close interaction between robots also introduces important safety challenges. During formation transitions, robots may move along intersecting trajectories, which increases the risk of inter-agent collisions. In addition, robots must also avoid surrounding obstacles while preserving the formation objective [6]. Therefore, a key challenge in multi-robot formation control is to design a control framework that can achieve the desired collective behavior while guaranteeing safety [7].

1.1 RELATED WORK

Recent years have witnessed significant advances in safe formation control for multi-agent systems. Classi-

cal methods often rely on path planning algorithms, such as Probabilistic Roadmaps and Rapidly-exploring Random Trees, to generate collision-free trajectories under given constraints [8]. Although these methods can provide feasible motion plans when a solution exists, their computational cost may limit their use in real-time applications. In addition, they often require prior knowledge of the environment, which is not always available in dynamic or partially unknown scenarios. Another common approach is the artificial potential field method [9], where obstacles or nearby agents generate repulsive effects to guide the robot away from unsafe regions. This approach is intuitive and relatively simple to implement. However, potential-field methods may suffer from local minima or deadlock issues, especially in dense multi-agent environments.

One popular alternative is to exploit Control Barrier Functions (CBFs) together with convex Quadratic Programming (QP) [10]. In this framework, CBFs encode safety requirements as constraints on the system states and control inputs. When these constraints are satisfied along the system trajectory, the safe set is rendered forward invariant, thereby providing formal safety guarantees. The resulting CBF constraints can be incorporated into a QP, whose objective is typically to keep the safe control input as close as possible to the nominal controller while enforcing safety. CBF-based methods have achieved significant success in multi-robot collision avoidance. For example, safety barrier certificates have been developed to guarantee collision-free behavior while minimally modifying nominal controllers [11]. More recent works have extended CBF-based safety control to practical objectives such as energy-efficient formation control [12], low-complexity CBF design for multi-agent systems [13], and distributed optimization-based safety verification [14].

However, many existing approaches still rely on continuous or time-triggered communication and control updates. This may lead to unnecessary communication overhead and increased computational burden, especially when the number of agents grows. As a result, improving communication efficiency while preserving formal safety guarantees remains an important challenge for scalable safe formation control.

Another promising problem - communication delays and model uncertainty in real systems

1.2 CONTRIBUTIONS

Motivated by these challenges, this project investigates a distributed safety-critical formation control framework combining distributed consensus control with a relaxed distributed Control Barrier Function formulation.

The main contributions of this work are:

- A distributed formation control framework is imple-

mented to drive agents toward the desired formation using local neighbor information.

- A relaxed distributed CBF-based (RDCBF) safety filter is integrated to guarantee inter-agent and obstacle collision avoidance.
- The proposed approach is evaluated through collision-prone simulation scenarios, including position swapping and formation control with static and dynamics obstacles.

As ongoing work, an event-triggered communication strategy is being developed to further reduce communication overhead. The objective is to design a triggering law that guarantees a lower-bounded inter-event time while preserving safety even when agents do not communicate continuously.

The remainder of this report is organized as follows. Section 2 presents the preliminary concepts of distributed formation control and Control Barrier Functions. Section 3 describes the proposed safety-critical formation control framework, including the collision avoidance CBF and the RDCBF formulation. Section 4 demonstrates the effectiveness of the proposed approach through simulation studies. Section 5 concludes the report and outlines future work. Additional theoretical derivations are provided in the appendices.

2 PRELIMINARIES

2.1 DISTRIBUTED FORMATION CONTROL

Consider a multi-agent system composed of N follower agents and one leader. The interaction between agents is represented by an undirected communication graph, where each agent exchanges information only with its neighbors. Let $x_i(t)$ denote the state of follower agent i , and $x_0(t)$ denote the leader state. The dynamics of each agent are modeled as a linear time-invariant system

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad (1)$$

where $u_i(t)$ is the control input.

In this project, we will use the double integrators dynamics model:

$$\begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0 & I_{2 \times 2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0 \\ I_{2 \times 2} \end{bmatrix} u_i \quad (2)$$

where $p_i \in \mathbb{R}^2$, $v_i \in \mathbb{R}^2$, and $u_i \in \mathbb{R}^2$ denote the position, velocity, and control input (acceleration command) of agent i , respectively. The velocity and acceleration of each agent are bounded by $\|v_i\|_\infty \leq \beta_i$ and $\|u_i\|_\infty \leq \alpha_i$.

The objective of formation control is to drive all follower agents toward a desired formation relative to the leader. Let f_i denote the desired relative position of agent i in the formation. The formation error is defined as

$$e_i(t) = x_i(t) - x_0(t) - f_i. \quad (3)$$

To achieve the desired formation, a distributed consensus-based control protocol is used. Then, the con-

trol input is designed as

$$\begin{aligned} u_i(t) &= Kz_i(t) + K'f_i, \\ z_i(t) &= \sum_{j \in \mathcal{N}_i} a_{ij} (x_j(t) - x_i(t) - (f_j - f_i)) \\ &\quad + d_i (x_0(t) - x_i(t) + f_i). \end{aligned} \quad (4)$$

where a_{ij} is the adjacency coefficient between agents i and j , d_i indicates whether agent i can access the leader information, and K, K' are feedback gain matrices.

This distributed controller only requires local neighbor information and guarantees formation convergence under a connected communication topology.

2.2 CONTROL BARRIER FUNCTION

Consider a control-affine system

$$\dot{x} = f(x) + g(x)u \quad (5)$$

where $f(x)$ is the drift term and $g(x)$ is the input matrix. Both f and g are assumed to be locally Lipschitz continuous. The state $x \in \mathcal{D} \subset \mathbb{R}^n$, where $\mathcal{D}$ denotes the state space of the system. The control input $u \in U \subset \mathbb{R}^m$, where U denotes the admissible control space.

Definition 2.1. A set $\mathcal{C}$ is called forward invariant if every trajectory starting in $\mathcal{C}$ remains in $\mathcal{C}$ for all future time, i.e.,

$$x(t_0) \in \mathcal{C} \Rightarrow x(t) \in \mathcal{C}, \quad \forall t \geq t_0. \quad (6)$$

where t_0 is an arbitrary initial moment.

Let the set $\mathcal{C} \subset \mathbb{R}^n$ denote the safe set, i.e., the set in which the robot states are required to remain. Then, the objective is to design a controller that guarantees the **forward invariance** of the set $\mathcal{C}$. To obtain a mathematically tractable formulation, the safe set $\mathcal{C}$ is represented as the **super-level set** of a CBF candidate $h : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}. \quad (7)$$

Definition 2.2. A continuous function $\kappa : (-b, a) \rightarrow \mathbb{R}$ for some $a, b > 0$ is called an extended class- $\mathcal{K}$ function if it is strictly increasing and satisfies $\kappa(0) = 0$.

Lemma 1. According to [10] and the proof in Appendix A, if the control input satisfies

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0, \quad (8)$$

where $\alpha(\cdot)$ is an extended class- $\mathcal{K}$ function, then $\mathcal{C}$ is forward invariant.

The time derivative of $h(x)$ along the trajectories of the system is given by

$$\dot{h}(x) = L_f h(x) + L_g h(x)u, \quad (9)$$

where $L_f h(x)$ and $L_g h(x)$ denote the Lie derivatives of h along the vector fields f and g , respectively. Then, given a CBF, the admissible control set $S(x) \subset U$ can be defined as

$$S(x) = \{u \in U \mid \dot{h}(x) + \alpha(h(x)) \geq 0, \quad \gamma > 0 \quad x \in \mathcal{D}\} \quad (10)$$

$$= \{u \in U \mid L_f h(x) + L_g h(x)u + \alpha(h(x)) \geq 0\}. \quad (11)$$

Therefore, the control objective is to design a CBF such that the above condition is satisfied.

For the extended class- $\mathcal{K}$ function $\alpha(\cdot)$, it regulates how quickly the system state can approach the boundary of $\mathcal{C}$. Different choices of $\alpha(\cdot)$, for example $\alpha(r) = r^P$ for any positive odd integer P , lead to different behaviors near the boundary. In this work, we chose

$$\alpha(h(x)) = \gamma h^3(x), \quad \gamma > 0, \quad (12)$$

leading to the constraint

$$\dot{h}(x) + \gamma h^3(x) \geq 0.$$

$$L_f h(x) + L_g h(x)u + \gamma h^3(x) \geq 0. \quad (13)$$

The choice of γ affects the performance of the CBF. A larger γ makes the system more conservative, meaning that the system reacts sooner when approaching dangerous scenarios. A smaller γ makes the system less conservative and the reaction to danger happens later.

Note that the linear time-invariant model belongs to the class of control-affine systems. Therefore, the dynamics model introduced in Section 2.1 can be used to construct the CBF constraints in the following methodology section.

3 METHODOLOGY

The proposed control architecture combines a distributed formation controller with a CBF-based safety layer as described in Figure 1.

The formation controller uses the desired relative positions and local neighbor information to compute a nominal control input $\hat{u}_i$, which drives each agent toward the desired formation, the applied control law is explained in 2.1.

Since this nominal input does not explicitly guarantee safety, it is passed through the CBF safety layer. To ensure that the agents are still able to finish the task controlled by the nominal controller, this layer modifies the control input only when necessary and keeps the modification minimal, producing a safe control input u_i^* for the agent dynamics. Based on this objective, the safety layer can be formulated as a QP optimization problem

$$\begin{aligned} u_i^* &= \arg \min_{u_i \in U_i} \|u_i - \hat{u}_i\|^2 \\ \text{s.t.} \quad & L_f h(x_i) + L_g h(x_i)u_i + \gamma h^3(x_i) \geq 0. \end{aligned} \quad (14)$$

For the communication topology, the distributed formation controller uses a static topology that guarantees formation convergence, while the safety layer employs a dynamic topology based on the sensing range of each agent.

3.1 CONTROL BARRIER FUNCTION FOR COLLISION AVOIDANCE

Intuitively, to ensure collision avoidance in a multi-robot system, we can consider safety pairwise rather than analyzing the whole system at once. Consider a pair of robots i and j , where i is the robot under consideration and j is one of its neighbors. To avoid collision, we determine whether the two robots can stop before reaching each other if they apply the maximum braking force at a given moment. Equivalently, the normal component of their relative velocity, denoted by $\Delta \bar{v}$, should be reduced to zero before collision occurs Figure 2.

Based on this idea, the safety condition is formulated as

$$\|\Delta p_{ij}\| + \int_{t_0}^{t_0+T_b} \Delta \bar{v}(t) dt \geq D_s, \quad \forall i \neq j. \quad (15)$$

where $\Delta p_{ij} = p_i - p_j$ is the relative position between agents i and j at time t_0 , t_0 is the time at which braking starts, T_b is the lookahead time such that $\Delta \bar{v}(t_0 + T_b) = 0$, and D_s is the minimum safety distance.

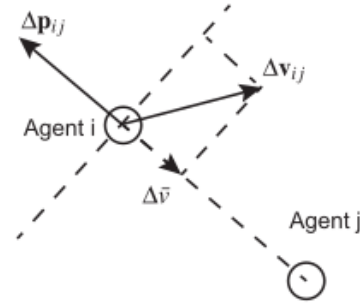


Figure 2: Relative position and velocity between two agents.

After some mathematical manipulation as stated in Appendix B. The obtained CBF is

$$h_{ij}(p, v) = \sqrt{2(\alpha_i + \alpha_j)(\|\Delta p_{ij}\| - D_s)} + \frac{\Delta p_{ij}^T}{\|\Delta p_{ij}\|} \Delta v_{ij}. \quad (16)$$

Then the pairwise safe set $\mathcal{C}_{ij}$ is defined as

$$\mathcal{C}_{ij} = \{(p_i, v_i) \in \mathbb{R}^4 \mid h_{ij}(p, v) \geq 0\}, \quad \forall i \neq j, \quad (17)$$

Substituting (16) into (13) and isolate u_{ij} , we obtain the following safety barrier constraint:

$$\begin{aligned} -\Delta p_{ij}^T \Delta u_{ij} &\leq \gamma h_{ij}^3 \|\Delta p_{ij}\| - \frac{(\Delta v_{ij}^T \Delta p_{ij})^2}{\|\Delta p_{ij}\|^2} + \|\Delta v_{ij}\|^2 \\ &\quad + \frac{(\alpha_i + \alpha_j) \Delta v_{ij}^T \Delta p_{ij}}{\sqrt{2(\alpha_i + \alpha_j)(\|\Delta p_{ij}\| - D_s)}}, \quad \forall i \neq j. \end{aligned} \quad (18)$$

where $\Delta u_{ij} = u_i - u_j$, and u_j is a constant.

This constraint will be used in all types of CBF-based collision avoidance safety layer, but it will be slightly modified to adapt to different control strategies (centralized and distributed).

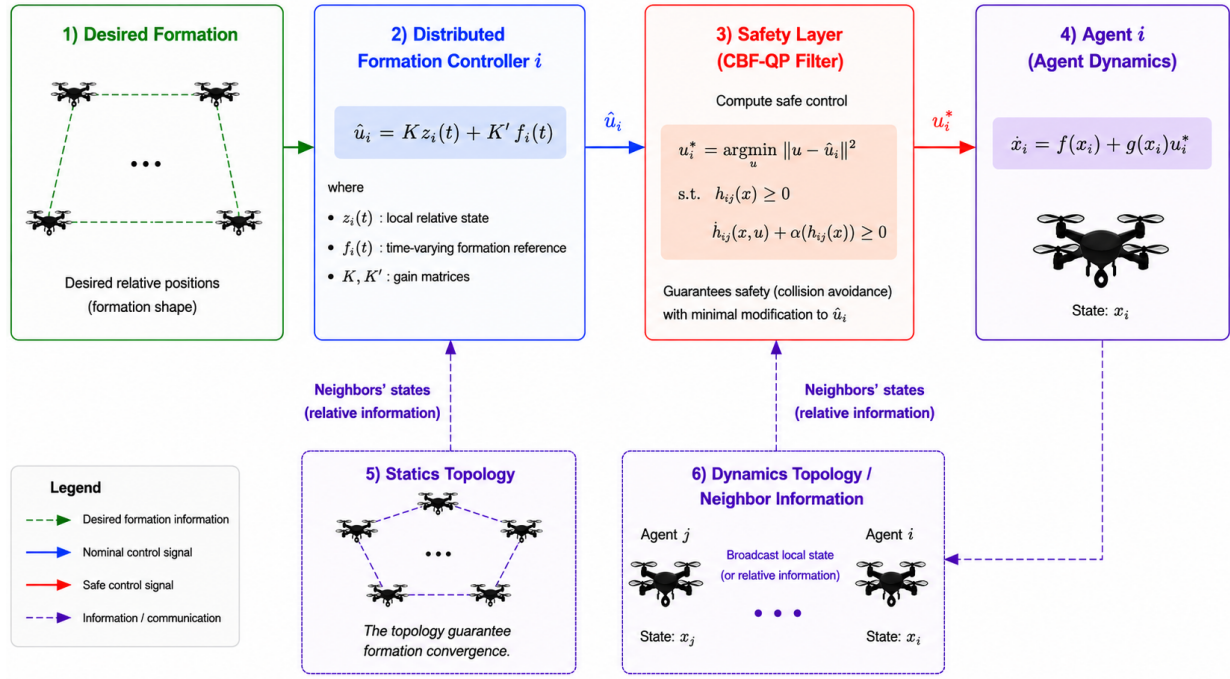


Figure 1: Proposed Control Structure.

Note that the proposed safety barrier framework can also be extended to static or moving obstacles. By modeling obstacles as circular regions, they can be treated similarly to a virtual agent with no control input. Consider an obstacle k with radius R_k centered at position p_k . A CBF can then be constructed to guarantee collision avoidance between agent i and obstacle k as

$$\bar{h}_{ik}(p, v) = \sqrt{2\alpha_i \left(\|\Delta p_{ik}\| - \left(\frac{D_s}{2} + R_k \right) \right)} + \frac{\Delta p_{ik}^T}{\|\Delta p_{ik}\|} \Delta v_{ik}, \quad (19)$$

where the obstacle is assumed to move with constant velocity v_k .

3.2 RELAXED DISTRIBUTED CBF

To implement this CBF, there are many types of CBF as centralized, distributed CBF but we use the relaxed distributed version, called RDCBF. The underlying reason is that the standard centralized CBF formulation can become computationally expensive as the number of agents increases. On the other hand, a purely distributed CBF may become too conservative, because the agent's admissible control space becomes smaller and smaller when more agents are involved.

For a RDCBF, instead of using the CBF constraint in Equation 13, we introduced a **relaxation factor** k_r , then the Relaxed CBF constrain become

$$L_f h(x) + L_g h(x)u + k_r(t)\gamma h^3(x) \geq 0. \quad (20)$$

Then for agent i , the pairwise safety barrier constraint 18 with agent j become

$$\begin{aligned} -\Delta p_{ij}^T \Delta u_{ij} &\leq k_{rj} \gamma h_{ij}^3 \|\Delta p_{ij}\| - \frac{(\Delta v_{ij}^T \Delta p_{ij})^2}{\|\Delta p_{ij}\|^2} + \|\Delta v_{ij}\|^2 \\ &+ \frac{(\alpha_i + \alpha_j) \Delta v_{ij}^T \Delta p_{ij}}{\sqrt{2(\alpha_i + \alpha_j)(\|\Delta p_{ij}\| - D_s)}}, \quad \forall i \neq j. \end{aligned} \quad (21)$$

In the nominal case, there is no relaxation on the safety barrier constraints, i.e., $\hat{K}_r = [\hat{k}_{r1}, \hat{k}_{r2}, \dots, \hat{k}_{rM}]^T = 1$ with $M = |\mathcal{N}_i|$ being the cardinality of agent i 's neighborhood set. The actual relaxation vector $K_r = [k_{r1}, k_{r2}, \dots, k_{rM}]^T$ is added to the cost of the QP-based controller to penalize deviation from $\hat{K}_r$.

$$\begin{aligned} (u_i^*, K_r^*) &= \arg \min_{u_i \in \mathbb{R}^2, K_r \in \mathbb{R}^M} J(u_i, K_r) = \|u_i - \hat{u}_i\|^2 \\ &+ c_K \|K_r - \hat{K}_r\|^2 \end{aligned} \quad (22)$$

$$\text{s.t.} \quad \tilde{A}_{ij} u_i \leq \frac{\alpha_i}{\alpha_i + \alpha_j} \tilde{b}_{ij}, \quad \forall j \in \mathcal{N}_i$$

$$\|u_i\|_\infty \leq \alpha_i$$

$$k_{rj} \geq 1 \quad \forall j \in \mathcal{N}_i$$

where c_K is the weight for the cost of relaxation, $\tilde{A}_{ij} = [-p_{ij}^T]$, and

$$\begin{aligned} \tilde{b}_{ij} = & k_{rj} \gamma h_{ij}^3 \|\Delta p_{ij}\| \\ & - \frac{(\Delta v_{ij}^T \Delta p_{ij})^2}{\|\Delta p_{ij}\|^2} + \|\Delta v_{ij}\|^2 \\ & + \frac{(\alpha_i + \alpha_j) \Delta v_{ij}^T \Delta p_{ij}}{\sqrt{2(\alpha_i + \alpha_j)(\|\Delta p_{ij}\| - D_s)}} \end{aligned} \quad (23)$$

Let the optimization parameter be

$$\mu_i = [u_i^T \quad \sqrt{c_K} K_r^T]^T \quad (24)$$

The QP-based controller can be rearranged as

$$\begin{aligned} \mu_i^* = & \arg \min_{\mu_i \in \mathbb{R}^{2+M}} J(\mu_i) = \|\mu_i - \hat{\mu}_i\|^2 \\ \text{s.t.} \quad & \bar{A}_{ij} \mu_i \leq \frac{\alpha_i}{\alpha_i + \alpha_j} \bar{b}_{ij}, \quad \forall j \in \mathcal{N}_i \\ & \|u_i\|_\infty \leq \alpha_i \end{aligned} \quad (25)$$

$$k_{rj} \geq 1 \quad \forall j \in \mathcal{N}_i$$

$$\text{where } \hat{\mu}_i = [\hat{u}_i^T \quad \sqrt{c_K} \hat{K}_r^T]^T$$

$$\bar{A}_{ij} = \left[-p_{ij}^T, 0, \dots, -\frac{\alpha_i \gamma h_{ij}^3 \|\Delta p_{ij}\|}{(\alpha_i + \alpha_j) \sqrt{c_K}}, \dots, 0 \right]$$

$$\begin{aligned} \bar{b}_{ij} = & -\frac{(\Delta v_{ij}^T \Delta p_{ij})^2}{\|\Delta p_{ij}\|^2} + \|\Delta v_{ij}\|^2 \\ & + \frac{(\alpha_i + \alpha_j) \Delta v_{ij}^T \Delta p_{ij}}{\sqrt{2(\alpha_i + \alpha_j)(\|\Delta p_{ij}\| - D_s)}} \end{aligned}$$

However, this QP-based controller might still face feasibility issues. Therefore we introduce a hybrid braking mode that provide alternative solutions when the infeasibility exists. The hybrid braking mode is that when the QP problem is feasible, the agent applies the optimal safe control input u_i^* . However, if the QP becomes infeasible, the agent can still remain safe by applying the maximum braking control until it stops.

Then, the control input is defined as

$$u_i = \begin{cases} u_i^*, & \text{if } u_i^* \text{ exists,} \\ -\alpha_i \frac{v_i}{\|v_i\|}, & \text{if } u_i^* \text{ does not exist and } \|v_i\| \neq 0, \\ 0, & \text{if } u_i^* \text{ does not exist and } \|v_i\| = 0. \end{cases} \quad (26)$$

Theorem 1. *If the robot agent i starts safe, the hybrid braking controller (26) guarantees that it always remains safe.*

Proof. When the QP-based controller (25) is feasible, agent i always stays safe as ensured by the CBF. When the QP-based controller becomes infeasible, agent i would apply the maximum braking force until it stops. Then it remains static until the QP-based controller becomes feasible again. Thus, agent i will stay safe regardless of the feasibility of (25). $\square$

4 SIMULATION RESULTS

To evaluate the effectiveness of the proposed approach, we design 2 test scenario where collision is highly likely to occur. The first one is *Position Swap*, the second one is *Hexagonal formation with static and moving obstacles*.

4.1 POSITION SWAP

Description: In this test, 19 agents are initially placed on a circle, and each agent is assigned a goal position located at the opposite point on the circle, as shown in Figure 3A. As shown in Figure 3B, when all agents move toward their respective targets, their trajectories naturally intersect near the center of the circle, creating a high likelihood of collisions.

Result: With the proposed approach, the robots avoid collisions while respecting the minimum safety distance, as shown in Figure 3C. Finally, they successfully reach their desired destinations, as shown in Figure 3D.

4.2 HEXAGONAL FORMATION WITH STATIC OBSTACLES

Description: In this test, 6 drones are required to form a hexagonal shape while avoiding 3 static obstacles. A disturbance region is also introduced to evaluate the robustness of the proposed approach. To simulate a more realistic scenario, the disturbance is modeled as wind acting on the drones. The wind direction is randomly drawn from a uniform distribution, while its amplitude is randomly chosen in the range $[0, 3]$ m/s².

Result: Figure 4 shows that the robots successfully avoid both the static obstacles and the disturbance region while moving toward the desired hexagonal formation. Despite the random wind disturbance, the proposed controller is still able to guide the robots to their target positions. Although the distributed formation controller and the CBF-based safety layer are not explicitly designed with a formal robustness mechanism, the simulation suggests a certain level of disturbance tolerance. Intuitively, the nominal formation controller continuously drives the agents back toward the desired formation, while the safety layer modifies the control input based on the observed state of the system. Therefore, even when the disturbance perturbs the trajectories, the safety layer can still react when agents approach unsafe regions and adjust the control input to maintain collision avoidance.

To further evaluate the performance of the proposed approach, Figure 5 presents the formation error of all agents. The formation error gradually converges to zero, indicating that the desired formation is successfully achieved. Figure 6 shows the minimum agent-agent distance and the minimum agent-obstacle distance during the mission. It can be observed that both distances remain above the prescribed safety threshold $D_s = 0.5$ at all times. Therefore, the proposed safety layer successfully guarantees collision avoidance while preserving the formation objective.

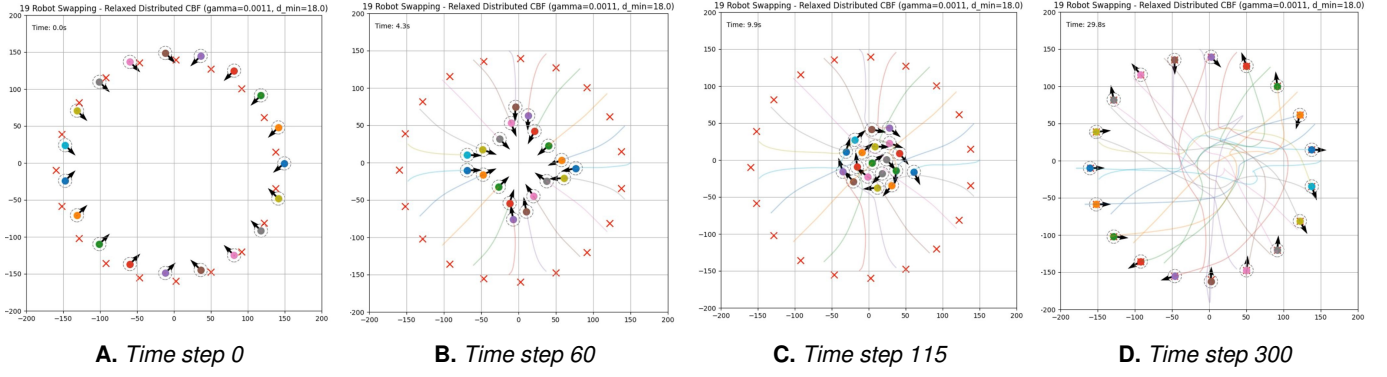


Figure 3: Simulation results of the 19-robot position-swapping task with $D_{\min} = 18$.

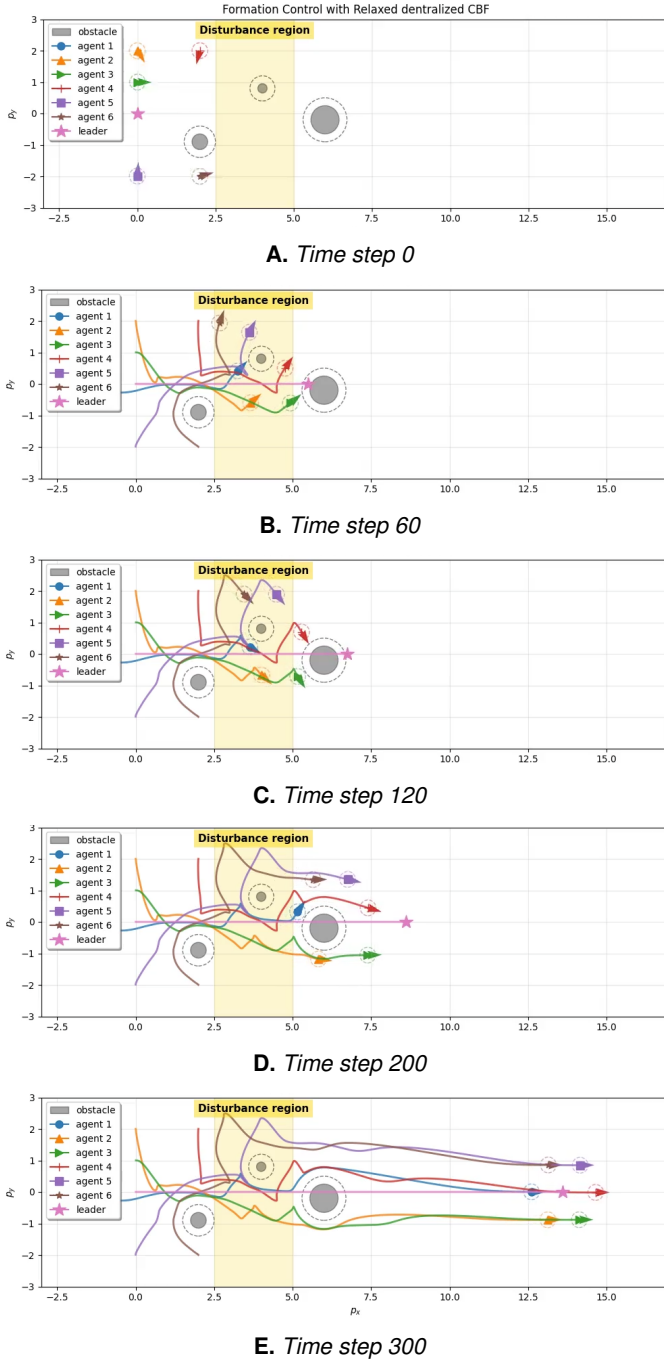


Figure 4: Simulation results of the hexagonal formation task with static obstacles and a disturbance region.

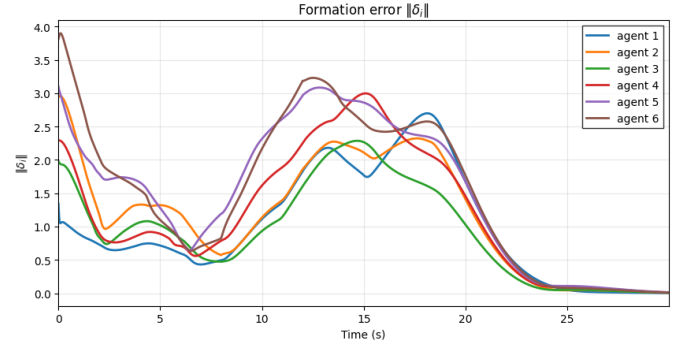


Figure 5: Formation error $\|\delta_i\|$ of all agents during the hexagonal formation task. The error converges to zero, indicating successful formation achievement.

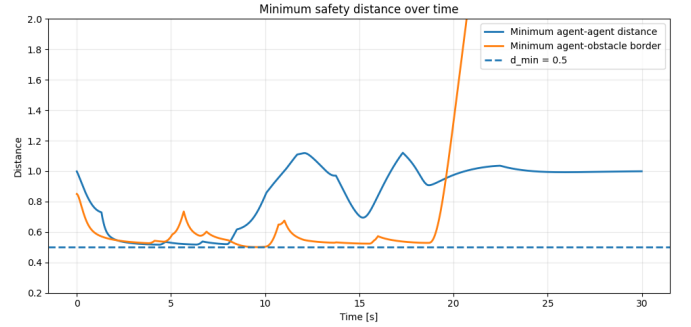


Figure 6: Minimum agent-agent and agent-obstacle distances over time. The safety distance threshold ($d_{\min} = 0.5$ m) is never violated throughout the mission.

5 CONCLUSION

This project presented a safe formation control framework for multi-robot systems by combining a distributed formation controller with a Relaxed Distributed Control Barrier Function (RDCBF)-based safety layer. The proposed approach enables robots to achieve the desired formation while guaranteeing collision avoidance with both neighboring agents and obstacles. Simulation results on collision-prone scenarios demonstrated that the proposed controller successfully maintains safety and achieves formation convergence, even in the presence of external disturbances. Future

work will focus on developing an event-triggered communication strategy to improve communication and computational efficiency while preserving the safety guarantees provided by the CBF framework.

A FORWARD INVARIANCE OF THE SAFE SET

Let the safe set be defined as

$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}. \quad (27)$$

If the control input satisfies

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0, \quad (28)$$

where $\alpha(\cdot)$ is an extended class- $\mathcal{K}$ function, then $\mathcal{C}$ is forward invariant.

Proof. Assume that the system trajectory starts inside the safe set, i.e.,

$$x(0) \in \mathcal{C}. \quad (29)$$

This implies

$$h(x(0)) \geq 0. \quad (30)$$

From the CBF condition,

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0, \quad (31)$$

we obtain

$$\dot{h}(x, u) \geq -\alpha(h(x)). \quad (32)$$

On the boundary of the safe set, we have

$$h(x) = 0. \quad (33)$$

Since

$$\alpha(0) = 0, \quad (34)$$

it follows that

$$\dot{h}(x, u) \geq 0. \quad (35)$$

Therefore, whenever the trajectory reaches the boundary of $\mathcal{C}$, it cannot evolve toward the unsafe region where

$$h(x) < 0. \quad (36)$$

Hence, every trajectory starting in $\mathcal{C}$ remains inside $\mathcal{C}$ for all future time. Therefore, $\mathcal{C}$ is forward invariant. $\square$

B DERIVATION OF THE COLLISION AVOIDANCE CBF

To avoid collision, the normal component of the relative velocity $\Delta \bar{v}$ should be reduced to zero before the agents reach the minimum safety distance.

The normal relative velocity is defined as

$$\Delta \bar{v} = \text{proj}_{\Delta p_{ij}} \Delta v_{ij} = \frac{\Delta p_{ij}^T}{\|\Delta p_{ij}\|} \Delta v_{ij}. \quad (37)$$

Assume that agents i and j apply the maximum relative braking acceleration. The time required to reduce $\Delta \bar{v}$ to zero is

$$T_b = \frac{0 - \Delta \bar{v}(t_0)}{\alpha_i + \alpha_j}. \quad (38)$$

The safety condition is then written as

$$\|\Delta p_{ij}\| + \int_{t_0}^{t_0+T_b} \Delta \bar{v}(t) dt \geq D_s. \quad (39)$$

Assuming constant braking acceleration,

$$\Delta \bar{v}(t) = \Delta \bar{v}(t_0) + (\alpha_i + \alpha_j)t. \quad (40)$$

Substituting this into the safety condition gives

$$\|\Delta p_{ij}\| + \int_{t_0}^{t_0+T_b} [\Delta \bar{v}(t_0) + (\alpha_i + \alpha_j)t] dt \geq D_s. \quad (41)$$

Then, we obtain

$$\|\Delta p_{ij}\| + \Delta \bar{v}T_b + \frac{\alpha_i + \alpha_j}{2}T_b^2 \geq D_s. \quad (42)$$

Substituting

$$T_b = \frac{-\Delta \bar{v}}{\alpha_i + \alpha_j}, \quad (43)$$

yields

$$\|\Delta p_{ij}\| - \frac{\Delta \bar{v}^2}{2(\alpha_i + \alpha_j)} \geq D_s. \quad (44)$$

Therefore,

$$\Delta \bar{v}^2 \leq 2(\alpha_i + \alpha_j) (\|\Delta p_{ij}\| - D_s). \quad (45)$$

Since collision can only occur when the agents move toward each other, we consider $\Delta \bar{v} \leq 0$. Hence,

$$\sqrt{2(\alpha_i + \alpha_j) (\|\Delta p_{ij}\| - D_s)} \geq -\Delta \bar{v}. \quad (46)$$

Finally, the CBF is defined as

$$h_{ij} = \sqrt{2(\alpha_i + \alpha_j) (\|\Delta p_{ij}\| - D_s)} + \frac{\Delta p_{ij}^T}{\|\Delta p_{ij}\|} \Delta v_{ij}. \quad (47)$$

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